Daniil Vlasenko

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Research Interests: Neuroscience, Network Science,
Data Analysis, Mathematical Modeling

Education

HSE University, Institute for Cognitive Neurosciences
PhD, Doctoral School of Cognitive Science
Moscow, Russia, November 2025 – present

HSE University, Institute for Cognitive Neurosciences
MS, Cognitive Sciences and Technologies
Moscow, Russia, September 2023 – June 2025

Saint Petersburg State University, School of Mathematics and Mechanics, Department of Statistical Modelling
BS, Applied Mathematics and Computer Science
St. Petersburg, Russia, September 2019 – June 2023

Additional Education

CNeuro, Theoretical and Computational Neuroscience Summer School
Beijing, China, 7-14 July, 2026

Neuromatch Academy, NeuroAI course
Online, 14 - 25 July 2025

Neuromatch Academy, Computational Neuroscience course
Online, 8 - 26 July 2024

Mediterranean School of Complex Networks
Grado, Italy, 30 June - 5 July 2024

Bioinformatics Institute, Professional retraining program "Algorithmic Bioinformatics"
St. Petersburg, Russia, September 2022 – January 2023

Selected Publications

(Q1 Scientific Journal) Krivonosov, M., Nazarenko, T., Ushakov, V., Vlasenko, D., Zakharov, D., Chen, S., Blyus, O., & Zaikin, A. (2025). Analysis of Multidimensional Clinical and Physiological Data with Synolitical Graph Neural Networks. Technologies, 13(1), 13.

Vlasenko D., Saranskaia I., Zakharov D., "From Pairwise to High-Order: Hypergraph Methods and Multivariate Connectivity Metrics for EEG/MEG," 2025 7th CNN, Nizhny Novgorod, Russian Federation, 2025, pp. 142-145.

Additional publications can be found on my Google Scholar page.

Selected Conferences

Volga Neuroscience Meeting 2025, The 7th International Conference "Neurotechnologies and Neurointerfaces", topic "From Pairwise to High-Order: Hypergraph Methods and Multivariate Connectivity Metrics for EEG/MEG" (poster presentation)
Nizhny Novgorod, Russia, 25 - 29 August 2025

International School and Conference on Network Science: **NetSci 2025**, topic "Ensemble-Based Graph Representation of fMRI Data for Cognitive Brain State Classification" (poster presentation)
Maastricht, the Netherlands, 2-6 June 2025

Baltic Forum 2024: Neuroscience, Artificial Intelligence and Complex Systems, VIII Scientific School "Dynamics of Complex Networks and their Applications", topic "*Ensemble methods for representation of fMRI, EEG/MEG data in graph form for classification of brain states*" (poster presentation).
Kaliningrad, Russia, 19–21 September 2024

Scholarships and Fellowships

Russian National PhD Research Scholarship, the program supporting PhD students and their dissertation research.
August2026 – present

Combined Master's-PhD track at HSE University, the talent program designed for students enrolled on full-tuition scholarships.
September 2023 – present

Academic Development Program (New Scientist category) at HSE University, the talent management program aimed at supporting the professional development of promising teachers and researchers.
January 2024 - May 2025

Honors and Awards

Winner (1st Place), Information Technology & Mathematics Section, 10th All-Russian Youth Scientific Forum "Science of the Future – Science of the Youth".
September 2025.

Certificate of Excellence "**Best Diploma of the Year 2025**", HSE University Institute for Cognitive Neuroscience.
June 2025

(Selected) Research experience

Agence Nationale de la Recherche project "DyndriteSymphony: Dynamics of nonlinear dendritic computation in cortical pyramidal neurons and interneurons"
External collaborator
September 2025 – present

My tasks: developing mathematical models of synaptic clustering that explain experimental observations, and investigating how passive dendrites can implement nonlinear computations.

Russian Science Foundation Interdisciplinary Grant 24-68-00030 "Next Generation Cognitive Artificial Intelligence"
Research-assistant
June 2024 - present

My tasks: developing graph and hypergraph representations of fMRI and EEG/MEG data, and applying network analysis and network-based machine learning to brain-state classification.

Teaching Experience

Supervision of course and diploma works of undergraduate students, **HSE University**
September 2024 – present

Seminar instructor (a few classes per course), "Advanced Computational Neuroscience", "Some Topics of Calculus, Differential Equations and Nonlinear Dynamics", **HSE University**
September 2024 – present

Languages

Russian (native), English (B2)

Skills

Programming
Python — NumPy, Pandas, Matplotlib, SciPy, Scikit-Learn, PyTorch, PyTorch Geometric, iGraph, NiLearn, MNE; R — dplyr, tidyr, ggplot2, iGraph; C++; MySQL; algorithms and data structures; Linux-based operating systems; familiar with remote server operation (terminal, Bash), version control systems (git, GitHub), HTML, CSS, JavaScript and Selenium; preparing documents and presentation slides using LaTeX.

Data analysis and machine learning
Classical data analysis; analysis of categorical data; network analysis; machine learning including deep learning; databases (SQL queries). Analysis and preprocessing of fMRI, EEG/MEG data; analysis of neural electrode data (neuropixels probes).

Mathematics
Statistics; probability theory; graph theory; linear algebra; mathematical analysis; (partial) differential equations; analytic geometry; optimization methods.